

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

M. Tech. Mechanical (CAD/CAM)

Sem-I University Examination, 2009-10

ME-704 ADVANCED MACHINE DESIGN

Date: 24/12/2009, Thursday

Time: 1:00 p.m. to 4:00 pm

Maximum Marks:70

Instructions:

1. Attempt **all** the questions.
2. Figures to the right indicate **full** marks.
3. Make suitable assumptions and draw neat figures wherever if required.
4. PSG Design data book is allowed in examination.

SECTION-I

Q1 (a) What do you mean by power rating of gears? Explain the power rating calculation as per BIS: 4460. **[05]**

Q1 (b) An edge crack beam, there is a crack in its central plane whose length is 5mm; a load of 2000N is applied opposite to crack, so that crack would tends to open in bending. Calculate the SIF of crack if the beam has the following dimensions: **[08]**

- Depth of the beam $w = 25\text{mm}$
- Thickness $b = 10\text{mm}$
- Span $S = 100\text{ m}$

If two loads of 600 N each are placed at 25mm from central plane. What SIF of crack be?

Q2 (a) Discuss on creep phenomena with necessary curve. **[04]**

Q2 (b) A pulley is keyed to a shaft midway between two bearings. The shaft is made of cold drawn steel for which the ultimate strength is 550 MPa and the yield strength is 400 MPa. The bending moment at the pulley varies from -150 Nm to +400 Nm. Determine the diameter of the shaft for indefinite life. The stress concentration factor for keyway in bending is 1.6. factor of safety is 1.5. size factor is 0.85 and surface finish factor is 0.90. **[08]**

OR

- Q2 (a)** A chemical reaction chamber working at a temperature of 723 K [09]
 uses steel bolts to tighten the two parts. The test on bolt material at
 this temperature resulted into strain rates of 3×10^{-18} per hr and
 2×10^{-18} per hr. If the bolts are tightened to a stress level 70 MPa
 initially. Calculate the time in which the stress level will be reduced to
 40 MPa. Assume that the chamber is made of same material and
 flanges are very rigid, modulus of elasticity at 723K is 1.7×10^5 MPa.
 Use following equations.

$$\epsilon_{cr} = C_2 \sigma^m \text{ and}$$

$$1/\sigma^{m-1} = 1/\sigma_i^{m-1} + EC_2 t / m$$

- Q2 (b)** Explain the safe life and fail safe design concepts. [03]
Q3 (a) Derive the equation for Soderberg criterion [03]
Q3 (b) Explain the importance of material selection and heat treatment in [03]
 gear design
Q3 (c) Derive correction factor with respect to corrective gear design [04]

OR

- Q3 (a)** Discuss on "damage-tolerant design". The latest design Philosophy [03]
 with the help of following plots (1) crack length v/s fatigue cycles (2)
 Fatigue crack growth v/s stress intensity factor range on log scale
Q3 (b) Explain different modes of fracture. [03]
Q3 (c) A 60 mm steel shaft transmitting 200 Nm torque is loaded by a [04]
 maximum bending moment of 100 Nm and an axial force of 5000 N.
 the material of shaft has yield strength 350 N/mm^2 . Determine the
 factor of safety of shaft by
- Maximum normal stress theory,
 - Maximum shear stress theory and
 - Distortion energy theory
 - Maximum strain energy theory .

SECTION-II

Q. 4(a) List and draw the various cross sections used for crane girder structures. Why hollow rectangular section is preferred? [06]

Q. 4(b) Explain the following terms. [03]

- Reliability,
- Maintainability,
- Availability

Q. 5(a) Explain Johnson's method of optimum design [04]

Q. 5(b) Design a tensile bar of length $L=200\text{mm}$ to carry a tensile load of 5 kN for minimum cost of the following materials: [09]

Material	Mass Density (kg/m^3)	Material Cost (Rs/N Weight)	Yield Strength (MPa)
A	7500	16	130
B	3000	32	50
C	4800	480	90
D	2100	32	20

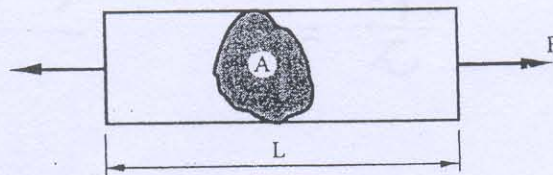
OR

Q. 5(a) Explain langrangian method for optimization [04]

Q. 5(b) Design a simple tensile bar of length L and load P with stiffness K_1 for cost optimization having constraints shown in figure below. [09]

$$K = K_1,$$

$$A \geq A_{\min}$$



Simple Tensile Bar

Q6 (a) Explain design procedure for the design of single main girder [04]

Q6 (b) Explain the testing of pressure vessel as per BIS. [04]

Q6 (c) Explain prestressing of thick cylinder with wire and tape wound. [05]

OR

Q6 (a) Explain various quadrants of legal and ethical domain. [04]

Q6 (b) Explain various codes of ethics. [04]

Q6 (c) Comparison of concurrent engineering and traditional engineering. [05]